higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(First Paper)
NQF LEVEL 3

4 NOVEMBER 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD	Reading from a table/graph/drawing
F	Choosing correct formula
SF	Substitution in formula
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off, etc.
R	Rounding off
U	Unit

This marking guideline consists of 10 pages

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QUESTION 1 [30] *Do not deduct marks if the 'R' or the '%' sign are omitted.		
Question	**Solution**	**Explanation**
1.1 1.1.1	=234 + 42✓ BODMAS = 276✓ (**answer only 0 marks**)	1 S +42 1 A (2)
1.1.2	= 0,5(5,3✓) - 2,35 = 2,65✓ - 2,35 = 0,3 ✓ (**answer only 0 marks**)	1 S 5,3 1 S 2,65 1 A (3)
1.1.3	8 + 16✓ = 24✓ (**answer only 0 marks**)	1 S (both correct) 1 A (2)
1.2	4 ✓: 2 ✓: 1✓	3 A (3)
1.3	R750 – R600 = R150✓ R150 ÷ R750✓ × 100% = 20%✓	1 M 1 Divide by R750 1 A (3)
1.4	$\frac{125}{1\,000}$✓ =0,125 ✓ kg (**Answer only full marks**)	1 M 1 C (2)
1.5	R2 700 ÷ 115✓ × 15✓ = R352,1739 =R352,17✓ OR R2 700 ÷ 115✓ x 100 = R2 347,83 R2 700 –R2 347,83✓ = R352,17✓	2 MA ÷115 and × 15 1 A (two decimal places) 1 M ÷115 1 MA subtracting 1 A (3)
1.6	285 ÷ 12✓ = 23,75✓ 23 full bags✓ (also accept 24 bags) (**Answer only full marks**)	1M ÷12 1 A 1R answer in context (3)

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1.7	1 : 500 000 23,7 : 23,7 × 500 000✓ = 11 850 000✓ cm = 118,5✓ km	1 × 500 000 1 A 1 C (correct answer) (3)

1.8	12:05:30 minus 11:20:00✓ 45 min 30 seconds✓ OR -1 hours +60 min 12:05:30 = 11 hours 65 min 30 seconds✓ Minus 11 hours 20 min 00 seconds 45 min 30 seconds✓ OR 11:20 to 12:00 = 40 min 12:00 to 12:05 = 5 min 45 min✓ + 30 seconds = 45 min 30 seconds✓	1 M 1 A (2)

1.9 1.9.1	5 × 500 = 2 500✓	1 MA (1)
1.9.2	R269,00 ÷ 2500✓ = R0,1076✓ ≈ R0,11 = 11 c✓	(CA 1.9.1) 1 M 1 A 1 R in cents (3)

QUESTION 2 [30] *Penalise once only for incorrect unit.		
Question	**Solution**	**Explanation**
2.1 2.1.1	C	1 A
2.1.2	D	1 A
2.1.3	B	1 A
2.1.4	F	1 A
2.1.5	A	1 A (5)

2.2 2.2.1	3 m ÷ 2 = 1,5 m✓	1 A (1)
2.2.2	Area = π r² = 3,14 × 1,5² ✓ = 7,065✓ ≈ 7,07 m²✓	1 SF (CA 2.2.1) 1 A 1 R (correct with unit) (3)

2.2.3	10,825 - 7,07✓ = 3,755✓ m²	1 M (CA 2.2.2) 1 A

	Also accept 3,76 m² OR 10,825 - 7,065✓ = 3,76 m²✓	(2)
2.2.4	Perimeter = 5 m + 5 m + 5 m✓ = 15✓ m	1 M 1 A (2)
2.3 2.3.1	V = π r² h = 3,14 × (150)² ✓× 744 ✓ = 52 563 600 ✓ mm³ = 52 563,6 cm³✓ OR V = π r² h = 3,14 × 15 × 15✓ × 74,4 ✓substitution ✓conversion = 52 563,6 cm³✓	1 A radius (150) 1 SF 1 A 1 C (÷ 1 000) OR 1 A radius (15) 1 C 1 SF 1 A (4)
2.3.2	52 563,6 ÷ 1 000 = 52,5636✓ = 53 litres✓ (**Answer only full marks**)	(CA 2.3.1) 1 C (52,5636) 1 A (2)
2.4 2.4.1	AB = 100 + 10 + 10✓ = 120✓ m (**Answer only full marks**)	1 M 1 A (2)
2.4.2	ABCD = 120✓ x 70✓ = 8 400 m² ✓	CA 2.4.1 2 SF 1 A (3)
2.4.3	SK = 100 ÷ 2 = 50 m ✓ (**Answer only full marks**)	1 A (1)

2.4.4	$KP^2 = PS^2 + SK^2$ $= 70^2 + 50^2$ $= 7\,400$✓ $\therefore KP = \sqrt{7\,400}$✓ $= 86,02$✓ $= 86$ m✓	CA 2.4.4 1 CA (7 400) 1 M $\sqrt{7\,400}$ 1 CA 1 R (4)

2.4.5	East✓	1 A (1)

QUESTION 3 [30] *Do not penalise if R is omitted.

Question	Solution	Explanation
3.1 3.1.1	16 July 2018✓ OR 16/07/2018 OR 16/7/2018	1 A (1)
3.1.2	20180716-241✓✓	1 A (1)
3.1.3 (a)	15 × R158✓ = R2 370✓ (**Answer only full marks**)	1 M 1 A (2)
(b)	R960 ÷ 3✓ = R320✓ (**Answer only full marks**)	1 M 1 A (2)
(c)	R6 529,96 × 15%✓ = R979,49✓ (**Answer only full marks**)	1 M 1 A (2)
(d)	R6 529,96 + R979,49✓ = R7 509,45✓ (**Answer only full marks**)	1 M 1 CA (Q3.1.3 c) (2)

3.2

BUDGET FOR 1st QUARTER 2019				
Item		Budget (Planned)	Actual (Realised)	Variance (Difference)
INCOME		**R**	**R**	**R**
Landscaping	1	45 000	35 000	- 10 000
Gardening Services	2	60 000	65 523	+ 5 523
Spaza Shop	3	90 000	98 435	**(e)** ✓ + **8 435**
TOTAL INCOME		**195 000**	**(b) 198 958**✓	**(f)** ✓ + **3 958**
EXPENSES		**R**	**R**	**R**
Rent	4	9 000	9 000	0
Wages	5	25 000	29 600	- 4 600
Water & Electricity	6	6 500	4 525	+ 1 975
Landline & Cell phone	7	3 600	(c) ✓**2 950**	+ 650
Vehicle fuel & maintenance	8	12 000	(d)✓**15 623**	- 3 623
Stationery & printer	9	2 500	1 550	(g) ✓**+ 950**
Other	10	3 000	325	+ **2 675**
TOTAL EXPENSES		**61 600**	**63 573**	(h) ✓ **- 1 973**
SURPLUS/DEFICIT		(a) **133 400**✓	**135 385**	**+ 1 985**

8 A (1 mark each per correct answer)
1 CA (SURPLUS/DEFICIT)

(9)

3.3. 3.3.1	R14 986,61✓✓	2 A (2)

3.3.2	PAY AS YOU EARN✓	1 A (1)

3.3.3	R 149,8661 × 1%✓ = R 149,87✓ (**Answer only full marks**)	1 M 1 A (2)

3.3.4	R14 986,61 – R1 012,96✓ = R13 97,65✓	1 M 1 CA (Q 3.3.1) (2)

3.3.5	5% = R5986,61 100% = x Sales = $\frac{\text{R5 986,61✓} \times 100✓}{5✓}$ = R11 9732,20✓ OR R5 986,61✓ ÷ 5✓ x 100✓ = R119 732,20✓	3 M 1 A 3 M 1 A (4)

QUESTION 4 [30]		
Question	**Solution**	**Explanation**
4.1 4.1.1	A = 4 x 15 ✓ = 60✓ B = 180 ÷ 120 ✓ = 1,5✓ (**Answer only full marks**)	 1 M 1 A 1 M 1 A (4)

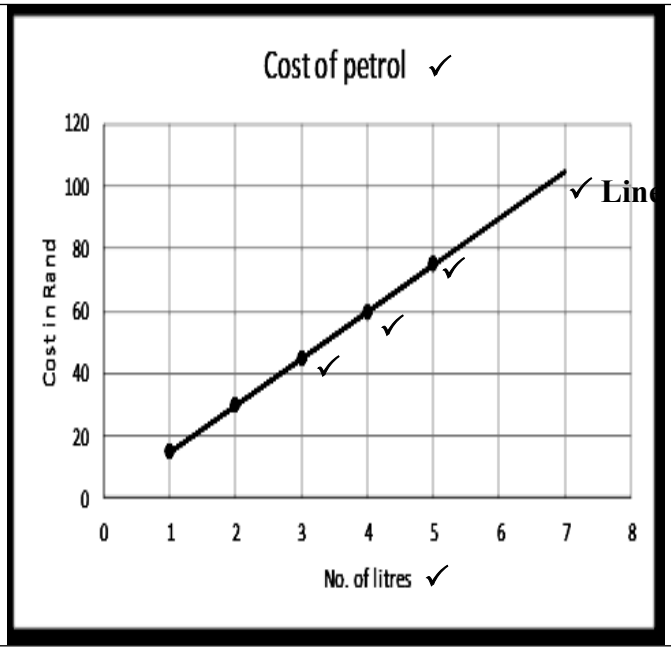
4.1.2	Cost of petrol = R15✓ × number of litres of petrol✓ **(accept other correct alternatives)**	2 A (2)

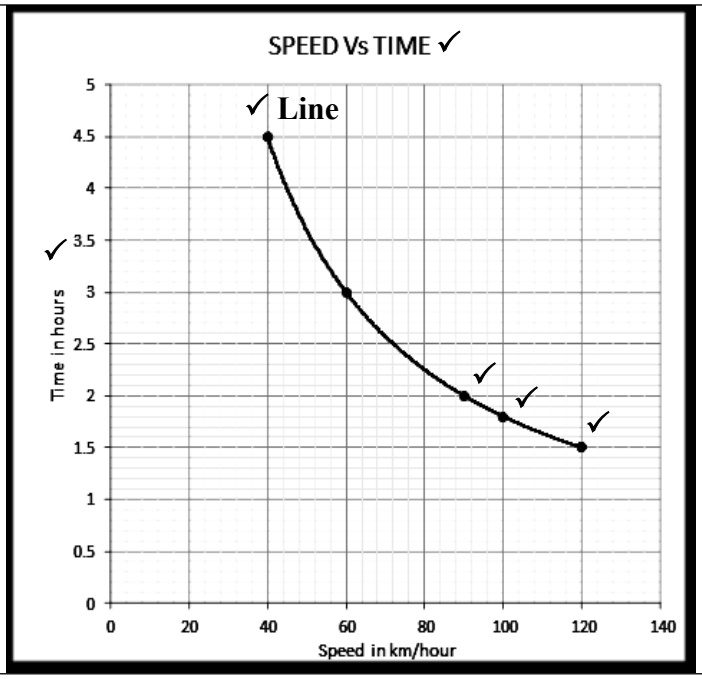
4.1.3	R255✓ ÷ R15✓ = 17✓ litres of petrol (**Answer only full marks**)	2 M 1 A (3)

4.1.4	Direct✓ relationship As the number of litres of petrol increases the cost of petrol increases✓	1 A 1 R/J (2)

4.1.5	indirect✓ relationship As the speed in increases the time taken decreases✓	1 A 1 R/J (2)

4.1.6	Speed in km✓ x time in hours✓ = 180✓ OR Speed in km ✓= 180✓ ÷ time in hours ✓ **(accept other correct alternatives)**	3A (3)

4.1.7		1 labelling title, 1 labelling both axes (no mark if one is omitted) 1 Straight Line 3A plotting any 3 points correctly (6)

4.1.8		1 labelling title, 1 labelling both axes (no mark if one is omitted) 1 Straight Line 3A plotting any 3 points correctly (6)

4.1.9	Time = 180 ÷ 150 = 1,2 hours = 1 hour✓ and 12 minutes✓	2 CA (2)

QUESTION 5 [30] *Do not penalise if R of % is omitted.

Question	Solution	Explanation
5.1 5.1.1	**FREQUENCY TABLE** **MEDIA** / **TALLY** / **FREQUENCY** Facebook / lll / 3✓ Twitter / ~~llll~~ lll / 8✓ Instagram / llll / 4✓ WhatsApp / lll / 3✓ e-Mail / lll / 3✓ TOTAL / [shaded] / **21**✓ Each tally and corresponding frequency must be correct for 1 mark per line	1 A for each correct line x 5 1 A TOTAL (6)
5.1.2	Twitter✓	1 RT (1)
5.2 5.2.1	Management✓✓	2 RT (2)
5.2.2	Sum = 105 + 100 + 190 + 70 + 100 + 40 + 110 = 715 Mean = 715✓ ÷ 7✓ = 102,14 = 102✓	2 MA 715 ÷ 7 1 R (no fraction for people) (3)
5.2.3	Sort: 40, 50, 145, **160,** 200, 250, 280✓ Median = 160✓ Management✓	1 A Sort 1A Median 1CA Programme (3)
5.2.4	Range = 190✓ – 40 ✓ = 150✓ (**Answer only full marks**)	2 RT 1 A (3)
5.2.5	Students enrolled = 715 + 1125✓ = 1 840✓ (**Answer only full marks**)	1 M 1 A (2)
5.3	The percentage increase adds up to 98%✓, it should add up to 100%.✓	2E (2)
5.4 5.4.1	Bar graph/compound bar graph✓	1 A (1)

5.4.2	Percetage increase $= \frac{60-58}{58}$✓$\times 100$✓ $= 3,45$✓%	2 M 1 A (3)

5.4.3	The prices on the vertical axis starts at R567✓ and not at zero. ✓ The prices on the vertical axis increases by only 50c. ✓ Thus causing the change in price to look bigger than it really is. ✓ The title of the chart has %✓ but the graph shows price increase in rands. ✓ **(any two of the above)**	2 R/J 2 R/J (4)

TOTAL: 150

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